

MULTIMODAL IMDB ANALYSIS WITH TENSORFLOW: EVALUATING CNN AND LSTM MODELS FOR FILM GENRE CLASSIFICATION

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INTRODUCTION

This report evaluates the implementation and performance of two machine learning models designed to classify films by genre using multimodal data from the Internet Movie Database (IMDB). Specifically, a Convolutional Neural Network (CNN) was trained to classify film posters by the genre, while a Long Short-Term Memory (LSTM) model was trained to classify film overviews by the genre. Both models were implemented using TensorFlow and Keras, leveraging GPU acceleration for training. The primary goal was to assess the models' effectiveness and critically evaluate their performance.

The assignment demonstrates the ability to design GPU-accelerated data pipelines, develop and train deep neural networks, and analyse their performance critically. This report details the data processing methods, model architectures, training processes, and evaluation results.

MODEL DESIGN AND TRAINING

Data Processing

The data processing phase involved preparing film posters and overviews for their respective models. Posters were resized to 64x64 pixels, normalized to $[0, 1]$, and processed through a TensorFlow pipeline leveraging batching, shuffling, and prefetching for efficiency. Overviews were tokenized using the TextVectorization layer, creating a vocabulary of the 10,000 most frequent words, and converted into padded sequences for LSTM input. GPU acceleration significantly reduced computational time across both pipelines.

Model Architectures and Training

The CNN, built with Keras Functional API, extracted spatial features through convolutional layers (filters: 16, 32, 64, 128), used max pooling to reduce dimensions, and dropout (0.2 and 0.5) to mitigate overfitting. Dense layers (1024 units) interpreted features, and a sigmoid output layer predicted 25 genres. It was compiled with the Adam optimizer (learning rate = $1e-4$), binary cross-entropy loss, and precision/recall metrics.

The LSTM, constructed with the Sequential API, included an embedding layer (dimension = 256), two bidirectional LSTM layers (256 and 128 units), a dense layer (128 units, ReLU activation), and dropout layers (0.5 and 0.8). The output layer used sigmoid activation for multi-label classification, and its compilation mirrored the CNN.

Both models used ModelCheckpoint to save the best weights based on validation loss and LearningRateScheduler for adaptive learning rates. The CNN was trained for 40 epochs, while the LSTM was trained for 20 epochs.

RESULTS AND ANALYSIS

CNN Model Performance

Loss and Precision: Training loss steadily declined, while validation loss fluctuated slightly but improved, stabilizing at approximately 0.27. Validation precision reached 0.65, indicating moderate accuracy for dominant genres.

Recall Trends: Recall stabilizes at around 0.28, indicating difficulty in detecting underrepresented genres, likely due to class imbalance.

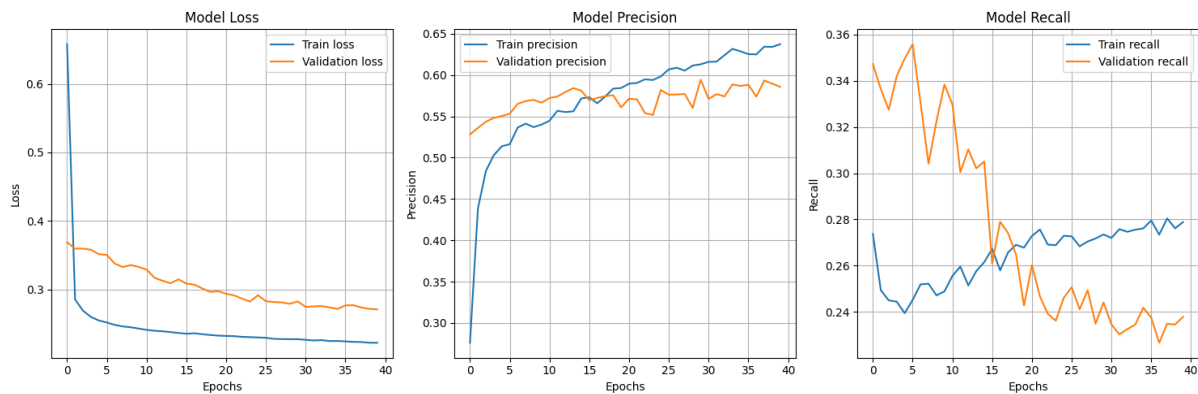


Fig 1: Plots depicting the progression of loss, precision, and recall over epochs for CNN, emphasizing learning trends and performance stability.

Critical Evaluation of CNN Model.

The CNN model demonstrates reliable convergence, with improving precision and reasonable generalization, as shown by its validation loss and precision trends. However, its low recall highlights difficulty in detecting underrepresented genres, primarily due to class imbalance. Addressing these challenges requires strategies like class weighting, oversampling, or focal loss to improve recall. Refining the learning rate schedule and increasing model capacity may further enhance performance. While effective for dominant genres, improving recall is essential for better handling of imbalanced datasets and achieving robust multi-label classification.

LSTM Model Performance

Loss and Precision: The training loss consistently declines, while validation loss stabilizes at 0.227, indicating effective learning and good generalization. Precision stabilizes around 0.60, showing reliable accuracy in predicting dominant genres.

Recall: Recall stabilizes at 0.23, reflecting challenges in identifying underrepresented genres due to dataset imbalance. Also, it highlights difficulty with rare genres, similar to the CNN.

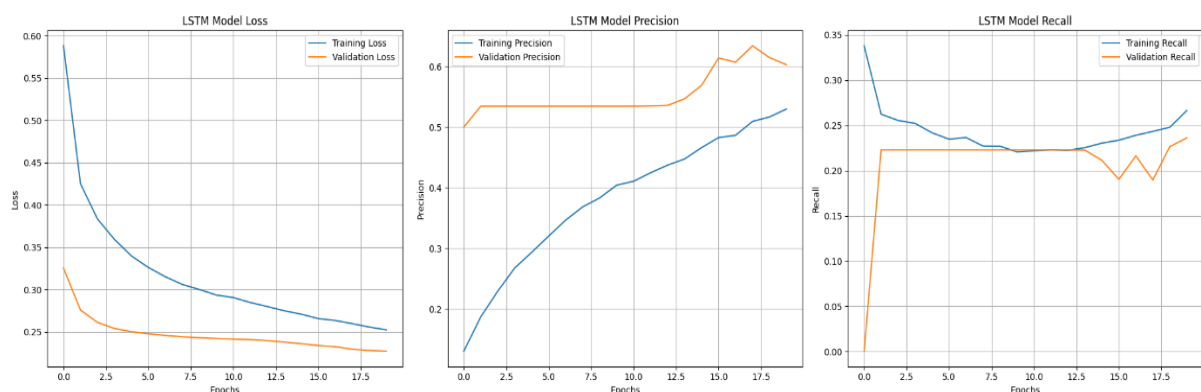


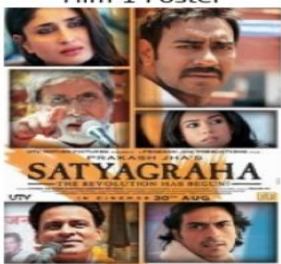
Fig 2: The plots display the evolution of loss, precision, and recall over epochs for LSTM, showing steady convergence and stable precision.

Critical Evaluation of LSTM Model.

The LSTM effectively captured sequential dependencies in text but faced challenges with underrepresented genres. Techniques such as weighted loss functions, oversampling, or integrating attention mechanisms could enhance recall. Despite its stable precision, addressing these limitations is crucial for better handling of imbalanced data.

Example predictions and critical evaluation of model performance across the classes.

Table 1: Comparison of Model Predictions with Ground Truth for Selected Films

Poster	Overview	Ground Truth Genres	Top Predicted Genres	Performance Insights
	A corporate go-getter sees his priorities shift when a personal tragedy brings him face-to-face with political corruption	Comedy, Drama	CNN: Drama, Action, Thriller LSTM: Drama, Comedy, Romance	The LSTM identified the comedic undertone better, reflecting its strength in understanding narrative context. The CNN leaned towards generic action elements.
	A magician competes in an international competition while falling in love with a street magician.	Comedy	CNN: Drama, Action, Thriller LSTM: Drama, Comedy, Romance	The LSTM successfully captured "Comedy," leveraging textual details, while the CNN misinterpreted visual cues as action-oriented genres.
	A group of seven former college friends gather for a week-end reunion at a South Carolina vacation home after the funeral of another of their college friends.	Horror, Mystery	CNN: Drama, Action, Thriller LSTM: Drama, Comedy, Romance	Both models struggled with suspenseful genres. The CNN focused on generalized predictions, while the LSTM misinterpreted the dramatic tone as comedic.

Insights from Predictions

Both models performed well for dominant genres like Drama but struggled with rare or overlapping genres. The LSTM's focus on narrative context helped it capture nuanced genres better than the CNN, which relied on visual patterns. These results highlight the importance of balanced datasets and specialized architectures.

Conclusion

This assignment demonstrated the use of TensorFlow and Keras to implement CNN and LSTM models for genre classification. While both models achieved moderate success, their performance was hindered by class imbalance and architectural limitations. Addressing these challenges through regularization techniques, attention mechanisms, and improved dataset curation could enhance model accuracy and generalizability.